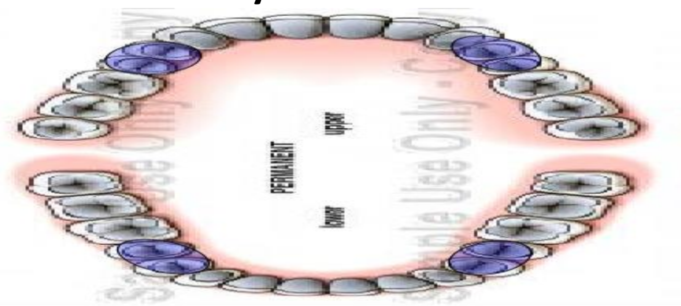
Premolars

4 maxillary & 4 mandibular



Beginning of calcification	Crown completed	Eruption	Root completed
$\frac{4}{4}$ 1.5 – 2.5 Y $\frac{5}{5}$ 2 – 2.5 Y	-5 YEARS	$\frac{10-11}{10-12}$ $\frac{10-12}{11-12}$	+ 3 YEARS

$\overline{4}$ & $\underline{4}$ Contact $\overline{3}$ & $\underline{3}$ → Mesially

Number of lobes (***4 lobes*** → **3** buccally and **1** lingually)

→ Middle lobe well developed → occlusally → Cusp

→ Buccally → buccal ridge

The lingual lobe developed in cusp

Upper 4

1- **Geometric outline on facial and lingual aspects**:

Trapezoidal → Short cervical

→ Long occlusal

2- **Outlines of crown** → Mesial and distal outlines and slopes → **concave.**

CA → mesially at junction of occlusal and middle third

CA → distally more occlusally

Mesial slope longer than distal slope

Buccal cusp pointed and long

→ Cervical outlines → convex root wise

3- **Outlines of root**
- **mesial and distal tapered to pointed apex & apex curved distal.**
- **Smooth and convex**

If two roots, the lingual root appears shorter and narrower than buccal root

Buccal:

S.A
- **Elevations** **convex with maximum convexity at cervical 1/3 & buccal rides**
- **Depressions** **shallow developmental depression mesial and distal to B ridge**

Lingual outline
- M & D outlines convex
- Lingual cusp shorter by 1 mm
- Distal slope of lingual cusp longer than mesial slope
- Cervical line convex root wise
- Convex with maximum convexity at m 1/3

Proximal aspects:

1- **Geometric outline** → trapezoidal
- long cervical
- Short occlusal

Significance of trapezoidal some triangular shape in proximal surface of interior teeth

2- **Outlines of crown**
- Buccal outline. Convex, maximum convexity at C 1/3
- Lingual Convex, maximum convexity M1/3
- Cervical line curves occlusally and decreased distally
- **Occlusal outline:**
  - B cusp longer than L cusp by 1 mm
  - Narrow occlusal table
  - B cusp tip below center of B root
  - L cusp tip online with L root
  - Mesial marginal rides Junction occlusal, M 1/3
  - Distal Marginal rides more occlusal

3- Surface anatomy of crown
- mesially
  - CA At junction of/M 1/3 OC Dimension
  - Slightly Buccally to midline BL dimension
- Distally → occlusally and more buccally

Mesial:

Mesial marginal developmental groove → Located lingual to CA & crossing MMR

Mesial developmental depression extends from cervically from CA then joins deep developmental depression between root bifurcation concave Fossa

Distal side → smooth and convex except small flat area cervical to CA

Roots:

In case of two roots (80%) → Mesial → root trunk half root length
→ Distal → root trunk longer as bifurcate Near apical 1/3

Smooth and convex except deep DD on root trunk → mesial

Smooth and convex except shallow on root trunk → Distal

In case of one root → Buccal and lingual outlines tapered to blunt apex
→ Smooth and convex except shallow depression In the center deeper mesially than distally

Occlusal aspect:

1- Geometric outline → hexagonal → Z Equal side buccal (MB, DB)
→ M Side shorter than D side
→ M Lingual shorter than DL side

* Thickness ›Width ‹ Crown wider buccally than lingually **why?**

Lingual convergence

2- S.A → Elevations → B Δ ridge & L Δ Ridge & M&D marginal ridge
Depressions → Central developmental groove & M&D Δ Fossae
Mesial marginal developmental Groove occlusal and mesial aspect show

Pulp cavity:

Bucco - lingual section → Wide pulp chamber & 2 pulp horn
→ 2 or one roots → 2root canal rarely 3 canals.

- Palatal root canal **wider** and more accessible than buccal

Mesial-distal section → Pulp chamber similar to upper canine but narrower
→ root canal shorter & narrow to apex.

Upper 5

$\bar{3}$ & $\underline{3}$ Contact $\bar{5}$ & $\underline{5}$ distally

1- Facial and lingual aspects have **trapezoidal** outline.

2- Facial outlines and surface anatomy of crown:

4	5
- Buccal cusp long and pointed.	- Buccal cusp short and blunt
- Longer M slope than D	- Shorter M slope than D
- Mesial CA junction OCC & M 1/3	- Mesial CA in occlusal 1/3
Distal more occlusal	Distal CA more cervically
- Cervical line curved root wise	- Cervical line less curved
- Prominent Buccal ridge	- Less prominent bridge
- Short root	- Long root

3- Lingual outlines and surface anatomy of crown

4	5
- Lingual cusp shorter by 1 mm Than buccal cusp	- L and B cusps at same level
- 80%, 2 roots	- Rare 2 roots (15%)
- Lingual root shorter than B root	- In case of z roots lingual root shorter

3 –**Proximal aspect→ trapezoidal→ small occlusal, long cervical Mesial aspect in:**

4	5
- Cusp longer than L cusp by 1 mm	- B&L cusps at some level
- Occlusal table narrow	- Occlusal table is wide.
- Mesial marginal DG and Connie fossa	- Crown surface smooth and convex Root has shallow developmental depression.
- Root track half root length	- MMR more occlusal
- MMR function occl & M 1/3	- CA ⟶ at acc 1/3 occlusal dim

4- Distal aspect

4	5
- DMR more occlusal than MMR - CA occlusally and more buccally than mesial CA - Smooth and convex surface expect area small flat cervical to CA. - Root trunk long as bifurcation near apical 1/3 - Surface smooth and convex except shallower DD on root trunk than mesially	- DMR more cervical than MMR - CA cervically and more buccally than MCA - Smooth and convex surface - One root and if 2 root bifurcation more apically - Surface smooth and convex except deeper DD in middle of root than mesially

5- Occlusal aspect

4	5
- It's hexagonal. - B & L Δ ridge - Long central developmental groove - Mesial marginal DG - M & D Δ fossa - No supplemental grooves	- It's oval. - B&l Δ ridges - Thicker M&D marginal ridge - Shorter central DG - Not found MM DG - M&D Δ fossa - Supplemental grooves

6- pulp cavity:

In case of single rooted tooth, mostly 1 root canal or may show single broad canal divided at mid-root into 2 canals by dentin islands, then the canals joined again near apical foramen & in case of 2 roots $\longrightarrow$ 2 canals one buccal and one lingual.

lower 4

→ 5 Aspects # → chronology #

1- Facial and lingual aspects:

A- Geometric outline → trapezoidal → small cervical (narrow)
→ Long occlusal (wide)

B- facial outlines of crown

→ Mesial and distal outlines and slopes concave

→ C.A → mesial at junction O/M thirds
→ Distal at cervically

Cervical line → convex root wise

- B cusp long and pointed.

Outlines of root → M, D outlines of root tapered to apex, curved distally

- Surface of root → smooth and convex

Similar 4, 5

- **Surface anatomy of crown:**
- **Elevation** → convex with maximum convexity at C 1/3 represent #
→ Middle lobe prominent buccally → buccal ridge
- **Depression** → shallow depression present mesial and distal to B ridge

C- Lingual outlines and surface anatomy of crown and root:

- M and D outlines convex

→ L cusp short and small reaching 2/3 crown length and pointed tip.

- Cervical line convex root wise
- **elevation** → lingual surface convex with maximum convexity at M 1/3

VIP → **Depression** → ML developmental groove at ML line angle

- Root tapered from cervical to pointed apex, curved distally.

D- Geometric outline of proximal aspect:

Rhomboid in shape with narrow occlusal table, lingual inculcation well prominent

- **significance of rhomboid outline:**
  1. Same 3 functions of trapezoidal in upper proximal post teeth
  2. Keep axis of mandibular and maxillary parallel.

- **If** mandibular crowns and roots have same relation as maxillary ones
  Cusps → hit and clash function and not proper inter cuspal relation.

E- Outlines of proximal aspect:

- Buccal outline convex and maximum convexity at C 1/3 represent#
- Lingual outline convex and maximum convexity at M 1/3
- Cervical line curves occlusally and decreased distally.
- Occlusal outline two cusps are not same level.

→ B cusp centered over the root due to lingual inclination.

→ L cusp tip in line with lingual outline of root

- DMR → straight and at right angle to the axis of tooth
- MMR → oblique inclined from B to L surface parallel to B cusp ridge.

M& D surfaces smooth and convex except cervical to CA concave

Distal CA → broader, cervically and lingual positioned

Mesial Aspect → ML DG at junction of L & M surfaces

Distal aspect → No ML DG

 # Outlines of root → B & L outlines are straight cervically then tapered apically to pointed apex

→ Mesial aspect → smooth and flat with DDG centered on root

→ Distal aspect more convex with shallow DD

F- Geometric outline of occlusal aspect diamond shape:

→ Lingual convergence → sharp

→ M-D (width) > B-L < thickness

- **# Elevation** → B Δ ridge → L Δ ridge small and short
  → M&D marginal ridges → # transverse ridge

  # depression → central DG → M& D fossa → M linear } snake
  → D circular } eyes appeared.

lower 5

5 aspects #   Chronology #

A- Geometric outline of crown facial and lingual:

wider trapezoidal **Due to** wider cervical third than lower 4

lower 5 → 3 cusp type < 5 lobes → 3B
→ 2 L

B- Facial outlines and surface anatomy:

4	5
- B cusp pointed and long well prominent B ridge. - CA more cervically	- B cusp short and less pointed less Prominent bridge. - CA broader and More occlusally

C- Lingual outline and surface anatomy:

4	5
- L cusp short and small reaching 2/3 crown length And has pointed tip. - Lingual surface convex and maximum Convexity at M 1/3 - ML DG at ML line angle	- 2 cusp type (lingual) - L cusp shorter and smaller than B cusp but larger than 4 - Surface convex and maximum convexity at o 1/3 - No ML DG

- 3 cup type → ML cusp longer and larger than DL cusp
  They both shorter than B cusp and less pointed
- → L D G Between 2 cusp

D- Proximal outlines → ***Rhomboid*** in shape with narrow occlusal table

4	5
- Lingual inclination well prominent - Maximum convexity at M 1/3 - L cusp short and small reaching 2/3 crown over the root - B cusp centered over the root. - MMR oblique & distal straight	- Lingual inclination less prominent - Maximum convexity at occ 1/3 - L cusp shorter and smaller than B cusp but larger than 4 - B cusp tip online with junction of B & M 1/3 of root - M& D MR straight

- Three cusp type
- ML cusp longer than DL cusp mesial aspect
- DL cusp shorter than ML cusp distal aspect

- surface anatomy similar to 4

E- Geometric outline of occlusal aspect:

Two cusp type

4	5
- Diamond in shape - Lingual convergence sharp	- Rounded in shape. - Slight lingual convergence

- # three cusp type → outline square / No lingual convergence

Elevations → 2-cusp type

- 4&5 → B& L Δ ridges / transverse ridge / M&D marginal ridges
- Dépressions → 4 → central DG & (M&D fossa) # < ML DG
  → 5 → central DG H&U shaped & M&D Δ Fossa & No ML DG

VIP →No supplemental groove in 4 & 5 present supplemental groove

→ 3 cup type → **Elevations** → B Δ ridges
→ L Δ ridges (ML & DL)
→ M&D marginal ridges

→ **Depressions** → CDG → Y shape
→ M& D Δ fossa
→ Central fossa slightly distal and lingual to midline
→ Supplemental groove present

F- Pulp cavity:

4	5
- Pulp cavity similar lower 3 But smaller in all dimension - Lingual pulp horn short **2 pulp horn**	- Pulp cavity larger - Lingual pulp horn smaller than B horn - Pulp chamber has 3 pulp horns in 3 cusp type • Larges B pulp horn → ML → DL